

Practical 2
4 February 2026

1. How many “Homo sapiens” sequences deposited in DDBJ? Compare with Genbank and EMBL
<http://www.ddbj.nig.ac.jp/>
<http://www.ncbi.nlm.nih.gov/genbank/>
<http://www.ebi.ac.uk/embl/>
2. What is the GC-content of the AY330867?
3. Compare the contents in DDBJ, Genbank and EMBL
4. Get the papers about “discrimination of beta barrel membrane proteins”.
<https://pubmed.ncbi.nlm.nih.gov/>
5. Find the papers published by any author (E.g.) Kihara D.
6. How many related articles are listed in PUBMED for the paper, Cell 2008 Dec 26;135(7):1158-9?
7. List the papers published in the journal “Nature” for the year 2024. Check the list in SCOPUS and PUBMED
8. Find the h-index and number of citations for “Burkhard Rost”.
9. Find the class of the enzyme EC 1.7.2.3 and its function
<http://www.brenda-enzymes.org/>
10. Find the catalytic site residues in Asparagine synthetase
Hint: Find the EC number and search in Catalytic site atlas
<https://www.ebi.ac.uk/thornton-srv/m-csa/>
11. Find the scientific name, the taxonomy ID and the number of chromosomes for the following organisms. <http://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi?mode=Root>
Organisms: Human, Cat, Dog, domestic guinea pig, and Thale cress
12. What are NCBI E-utilities? Give the syntax for fetching a record in FASTA format using E-utilities. <http://www.ncbi.nlm.nih.gov/books/NBK25500/>
13. List two databases under each of the following category.
 - a. Protein properties, b. Small molecules (Structure related), c. Cancer gene databases

Hint: Use Nucleic Acids Research (NAR) – ‘database category list’
https://www.oxfordjournals.org/our_journals/nar/database/c/

Deadline: 10 Feb 2026